Kunwoo Park

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Research Interests

Integrated photonics; Disordered photonics; Topological photonics.

Education

Integrated M.S./Ph.D., Department of Electrical and Computer Engineering — 2024.09 – present
Seoul National University Advisor: Sunkyu Yu — *Expected: 2029.08*

B.S., Department of Electrical and Computer Engineering — 2018.03 – 2024.08
Seoul National University

Military Service, Republic of Korea Army — 2020.12 – 2022.06

Research Experience

- **Integrated Photonics.** Design and experimental demonstration of photonic circuits, with theoretical work toward all-optical neural networks.
- **Disordered Photonics.** Development of hypergraph models of wave scattering for scalable design of disordered photonic materials.
- **Topological Photonics.** Investigation of non-Abelian topology for robust and programmable control of light.

Publications

Preprints

1. Jungyoon Kim†, Kyuho Kim†, **Kunwoo Park**, Namkyoo Park* & Sunkyu Yu*. Task-specific programming of chaos in neural circuits.
   arXiv:2605.19465
2. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Hypergraph modelling of wave scattering to speed-up material design.
   arXiv:2507.15329

Journal Articles

1. Gitae Lee†, Seungmok Youn†, Ikbeom Lee†, **Kunwoo Park**, Duhwan Hwang, Seungkyun Park, Xianji Piao*, Namkyoo Park* & Sunkyu Yu*. Non-Hermitian stealthy hyperuniformity.
   Advanced Science 13, e77054 (2026)
2. Seungmok Youn, **Kunwoo Park**, Ikbeom Lee, Gitae Lee, Namkyoo Park* & Sunkyu Yu*. Phase-sensitive engineering of optical disordered materials using heterogeneous networks.
   Advanced Optical Materials 14, e03499 (2026)
3. Kyuho Kim†, **Kunwoo Park**†, Hyungchul Park, Sunkyu Yu, Namkyoo Park* & Xianji Piao*. Programmable photonic unitary circuits for light computing.
   Nanophotonics 14, 1429 (2025)

International Conferences

1. Ikbeom Lee, **Kunwoo Park**, Seungmok Youn & Sunkyu Yu*. Inverse design of microstructures of non-Hermitian scattering systems.
   SPIE Optics + Photonics 2025, 13578-59, San Diego, California, USA, 6 August 2025
2. **Kunwoo Park**, Ikbeom Lee, Seungmok Youn & Sunkyu Yu*. Modularization of light scattering for photonic computing applications.
   SPIE Optics + Photonics 2025, 13581-25, San Diego, California, USA, 5 August 2025
3. Ikbeom Lee, **Kunwoo Park** & Sunkyu Yu*. Design of wave scattering from gain and loss disordered materials.
   CLEO-PacificRim (CLEO-PR) 2024, Mo2J-4, Incheon, Republic of Korea, 5 August 2024
4. **Kunwoo Park**, Ikbeom Lee & Sunkyu Yu*. Inverse design method of hyperuniform materials using system factorization.
   CLEO-PacificRim (CLEO-PR) 2024, Mo2J-2, Incheon, Republic of Korea, 5 August 2024

Domestic Conferences

1. **Kunwoo Park**, Gitae Lee, Ikbeom Lee, Seungmok Youn, Namkyoo Park & Sunkyu Yu. Large-Scale Optical Material Design via Hypergraph Models.
   Optics and Photonics Congress 2026, W3C-V.01, Jeju, Republic of Korea, 22 July 2026.
2. Ikbeom Lee, Gitae Lee, Seungmok Youn, **Kunwoo Park**, Duhwan Hwang, Xianji Piao, Namkyoo Park & Sunkyu Yu. Designing asymmetric scattering using gain and loss media.
   Optics and Photonics Congress 2026, TP-V.23, Jeju, Republic of Korea, 21 July 2026.
3. Duhwan Hwang, Jongheon Yeo, **Kunwoo Park**, Namkyoo Park & Sunkyu Yu. Numerical Analysis of Single-Photon Dynamics in Nonreciprocal Waveguide-Resonator Coupled Systems.
   Photonics Conference 2025, T2A-II.05, Pyeongchang, Republic of Korea, 27 November 2025.

Patents

1. Nonlinear unitary optical device.
   Sunkyu Yu, Namkyoo Park, Xianji Piao & **Kunwoo Park**
   KR 10-2859166 (Application No. 10-2025-0005056; Provisional No. 10-2024-0179951)
   Registration date: 2025.09.09; Filing date: 2025.01.13; Provisional filing date: 2024.12.05
   US application: 19/048,370; Filing date: 2025.02.07
   EU application: 25156862.2; Filing date: 2025.02.10

Grants and Fellowships

Graduate Research Encouragement Grant (석사과정생연구장려금지원사업) 2025.09 – 2026.08
Principal Investigator

Teaching

Introduction to Electromagnetism with Practice (Undergraduate), Teaching Assistant Fall 2026 (ongoing)
Seoul National University

Application of Quantum Mechanics (Graduate), Teaching Assistant Spring 2025
Seoul National University

Topics in Electro-physics (Graduate), Teaching Assistant Spring 2025
Seoul National University

References

Available upon request.

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